

Practice Session: GPIO, Buttons & LEDs

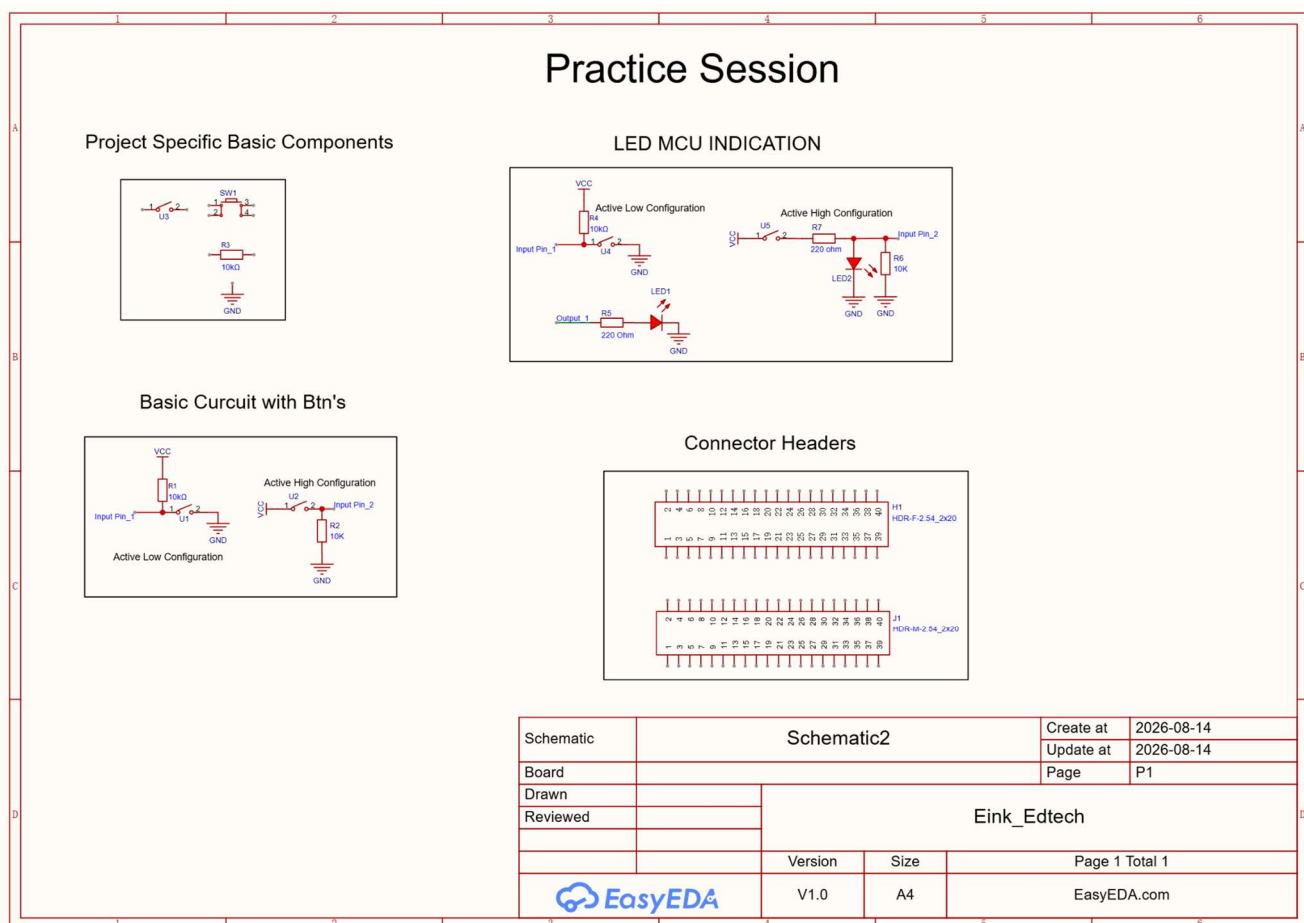
A beginner-friendly introduction to reading simple MCU circuits

1. What We Will Learn

- Identify common components used in a small MCU project.
- Understand how a push button can be used as a digital input.
- Understand the difference between Active LOW and Active HIGH.
- Understand how an LED is connected to a digital output.
- Recognize the role of pull-up, pull-down, and current-limiting resistors.
- Relate the schematic to the physical connector/header pins.

2. Practice Circuit Reference

The original practice schematic is shown below. You do not need to understand every symbol immediately. Use the sections that follow to learn what each part means.



3. Know Your Basic Components

Component	What it does	Remember
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Push button / switch	Connects or disconnects two points when pressed.	Think: ON / OFF input.
10 kΩ resistor	Helps keep a GPIO input at a known HIGH or LOW level.	Usually used as a pull-up or pull-down.
220 Ω resistor	Limits current flowing through an LED.	Protects the LED.
LED	Lights when current flows through it in the correct direction.	Used as a simple output indicator.
GND	The circuit's common reference point.	Many circuit signals are measured relative to GND.
VCC	The circuit's positive supply voltage.	The actual voltage depends on the MCU/project.

4. Reading a Simple Schematic

A schematic is a map of an electronic circuit. Lines show electrical connections, while symbols represent components.

- VCC → positive supply
- GND → ground/reference
- GPIO / Input Pin → a pin the MCU reads
- GPIO / Output Pin → a pin the MCU controls
- Resistor → limits current or fixes a signal level
- Switch → changes whether two points are connected

5. Push Buttons as Digital Inputs

A microcontroller input should normally have a clear digital state: HIGH or LOW. A resistor is used so the input does not 'float' when the button is not pressed.

Active LOW

In an Active LOW circuit, pressing the button connects the input to GND. Therefore, the MCU reads LOW when the button is pressed.

Button	GPIO level	Meaning
Released	HIGH	Button is OFF
Pressed	LOW	Button is ON

Easy way to remember: Pressed → LOW.

Active HIGH

In an Active HIGH circuit, pressing the button connects the input to VCC. Therefore, the MCU reads HIGH when the button is pressed.

Button	GPIO level	Meaning
Released	LOW	Button is OFF
Pressed	HIGH	Button is ON

Easy way to remember: Pressed → HIGH.

6. Why Do We Need the 10 kΩ Resistor?

When a button is not pressed, the GPIO input can otherwise be left without a definite HIGH or LOW value. This is called a floating input. A pull-up or pull-down resistor gives the input a default state.

Type	Default state	Button changes it to
Pull-up	HIGH	LOW when connected to GND
Pull-down	LOW	HIGH when connected to VCC

7. Using an LED as a Digital Output

An LED can show us what the MCU is doing. The MCU changes an output pin HIGH or LOW, and the LED responds.

The typical beginner circuit is:

MCU Output → 220 Ω Resistor → LED → GND

The 220 Ω resistor limits the current through the LED. Do not connect a typical LED directly to a GPIO pin without appropriate current limiting.

MCU output	LED	What you see
HIGH	ON	LED lights
LOW	OFF	LED is off

8. Connector Headers & Physical Pins

The connector header is the bridge between the schematic and the physical MCU board. Each numbered pin represents a real electrical connection.

- Find the board's pinout before connecting wires.
- Do not assume that every numbered pin is a GPIO.
- Some pins provide power (VCC), some are GND, and others are GPIO or special-purpose pins.
- Always check which voltage the MCU uses before connecting VCC.

9. Practice Challenge

Build a simple circuit in which pressing a button causes an LED to turn ON.

1. Identify the button, LED, 10 kΩ resistor, 220 Ω resistor, VCC, and GND.
2. Choose one MCU pin as a digital input for the button.
3. Choose one MCU pin as a digital output for the LED.
4. Connect the button using either the Active LOW or Active HIGH configuration.
5. Connect the LED through the 220 Ω resistor.
6. Write or use a simple program that reads the button and controls the LED.
7. Test what happens when the button is pressed and released.

10. Quick Check

What is the difference between an input pin and an output pin?

Answer: _____

In Active LOW, what does the MCU read when the button is pressed?

Answer: _____

In Active HIGH, what does the MCU read when the button is pressed?

Answer: _____

Why is a 10 k Ω pull-up or pull-down resistor used?

Answer: _____

Why is a resistor placed in series with the LED?

Answer: _____

What does GND mean?

Answer: _____

Why should you check the MCU pinout before wiring the header?

Answer: _____

11. Remember These 5 Things

- GPIO INPUT → the MCU reads a signal.
- GPIO OUTPUT → the MCU controls a signal.
- Active LOW → pressed usually means LOW.
- Active HIGH → pressed usually means HIGH.
- Resistors help control current or give inputs a known default state.

Start simple. Understand the signal first, then build the circuit.